

L^AT_EX 数学公式对齐

L^AT_EX 的宏包 `amsmath` 提供了很多除 L^AT_EX 自带如 `$$公式$$`、`\[公式\]` 之外的行间公式环境，本文将如何排版多行公式对齐为主要内容，分享典型的应用案例。

<code>equation</code>	<code>equation*</code>	<code>align</code>	<code>align*</code>
<code>gather</code>	<code>gather*</code>	<code>alignat</code>	<code>alignat*</code>
<code>multline</code>	<code>multline*</code>	<code>flalign</code>	<code>flalign*</code>
<code>split</code>	<code>cases</code>		

通常而言，上面的公式环境中带 “*” 号一般是去掉公式编号，而不带 “*” 号的公式通常会自动的进行编号。而在公式对齐方面我们可以分为多行对齐，多列对齐，多行多列对齐以及左中右对齐等。

一、长公式折成多行对齐

多行公式可分为同一公式折行对齐和多行公式对齐，前者一般都会选择如 `=`、`>`、`<`、`≤`、`≥` 位置对齐，而后者通常会选择同一运算符或者左对齐方式。同一公式折行对齐如：

```
\begin{align*}
f^{\prime}(x)
&= \int f^{\prime\prime}(x) \mathrm{d} x \\
&= \frac{\mathrm{d}}{\mathrm{d} x} f(x) \\
&= \frac{\mathrm{d}^2}{\mathrm{d} x^2} \int f(x) \mathrm{d} x \\
&= \frac{\partial}{\partial x} f(x) \mathrm{d} x
\end{align*}
```

$$\begin{aligned} f'(x) &= \int f''(x) \mathrm{d} x \\ &= \frac{\mathrm{d}}{\mathrm{d} x} f(x) \\ &= \frac{\mathrm{d}^2}{\mathrm{d} x^2} \int f(x) \mathrm{d} x \\ &= \frac{\partial}{\partial x} f(x) \mathrm{d} x \end{aligned}$$

```
\begin{align*}
(Z_2 - Z_1) &= -\frac{R_d}{g_0} \bar{T}_v \ln \left( \frac{dp}{p} \right) \\
&= -\frac{287 \times 255}{9.81} \ln \left( \frac{P_2}{P_1} \right) \\
&= -7.4 \ln \left( \frac{P_2}{P_1} \right)
\end{align*}
```

$$\begin{aligned} (Z_2 - Z_1) &= -\frac{R_d}{g_0} \bar{T}_v \ln \left(\frac{dp}{p} \right) \\ &= -\frac{287 \times 255}{9.81} \ln \frac{P_2}{P_1} \\ &= -7.4 \ln \frac{P_2}{P_1} \end{aligned}$$

```
\begin{align*}
\int_0^{\infty} \mathrm{d} \frac{\sin(x)}{x} \mathrm{d} \frac{\sin}{\left( \frac{x}{3} \right)} \mathrm{d} \frac{\sin}{\left( \frac{x}{15} \right)} \mathrm{d} \frac{\sin}{\left( \frac{x}{15} \right)} \mathrm{d} x \\
&= \frac{467807924713440738696537864469}{935615849440640907310521750000} \pi
\end{align*}
```

```
&=\dfrac{\pi}{2}-\dfrac{6879714958723010}{935615849440640907310521750000}\pi\\
&=\dfrac{\pi}{2}-2.31\times 10^{-11}\\
\end{align*}
```

$$\int_0^\infty \frac{\sin(x)}{x} \frac{\sin\left(\frac{x}{3}\right)}{\frac{x}{3}} \cdots \frac{\sin\left(\frac{x}{15}\right)}{\frac{x}{15}} dx = \frac{467807924713440738696537864469}{935615849440640907310521750000} \pi$$

$$= \frac{\pi}{2} - \frac{6879714958723010}{935615849440640907310521750000} \pi$$

$$= \frac{\pi}{2} - 2.31 \times 10^{-11}$$

二、多个公式对齐

对于多个式子长度相当，且有相同或明显相似运算符的多个式子，我们通常选择在该运算符前对齐。

```
\begin{align*}
\int_0^\infty \dfrac{\cos x}{1+x^2} dx & \\
&=\frac{\pi}{2e} \\
\int_0^\infty \dfrac{\cos xy}{\left(1+x^2\right)} dx &=\dfrac{\pi}{e^{-y}} \\
&e^{\{-y\}} \\
\int_0^\infty \dfrac{\cos xy}{x^2+a^2} dx & \\
&=\frac{\pi}{e^{-ay}} \\
&e^{\{-ay\}} \\
\end{align*}
```

$$\int_0^\infty \frac{\cos x}{1+x^2} dx = \frac{\pi}{2e}$$

$$\int_0^\infty \frac{\cos xy}{(1+x^2)} dx = \frac{\pi e^{-y}}{2}$$

$$\int_0^\infty \frac{\cos xy}{x^2+a^2} dx = \frac{\pi e^{-ay}}{2a}$$

```
\begin{align*}
d\dot{q} &=d\dot{u}+d\dot{w} \\
\frac{DU}{Dt}&=\frac{Dq}{Dt}-\frac{Dw}{Dt}\\
C_v\frac{DT}{Dt}&=\frac{Dq}{Dt}-p\frac{D\alpha}{Dt} \\
\end{align*}
```

$$d\dot{q} = d\dot{u} + d\dot{w}$$

$$\frac{DU}{Dt} = \frac{Dq}{Dt} - \frac{Dw}{Dt}$$

$$C_v \frac{DT}{Dt} = \frac{Dq}{Dt} - p \frac{D\alpha}{Dt}$$

```

\begin{align*}
d\Big(\frac{\partial f}{\partial x}\Big)
&= \frac{\partial^2 f}{\partial x^2} dx + \frac{\partial^2 f}{\partial x \partial y} dy \\
d\Big(\frac{\partial f}{\partial y}\Big)
&= \frac{\partial^2 f}{\partial x \partial y} dx + \frac{\partial^2 f}{\partial y^2} dy
\end{align*}

```

$$d\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial x^2} dx + \frac{\partial^2 f}{\partial x \partial y} dy$$

$$d\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial x \partial y} dx + \frac{\partial^2 f}{\partial y^2} dy$$

```

\begin{align*}
\frac{df}{dx}\Big|_{x=x_0} &\simeq \frac{f(x + \Delta x, t) - f(x, t)}{\Delta x} + O(\Delta x) \\
\frac{df}{dx}\Big|_{x=x_0} &\simeq \left[ \frac{f(x + \Delta x, t) - f(x, t)}{\Delta x} \right] + O(\Delta x)
\end{align*}

```

$$\frac{df}{dx}\Big|_{x=x_0} \simeq \left[\frac{f(x + \Delta x, t) - f(x, t)}{\Delta x} \right] + O(\Delta x)$$

$$\frac{df}{dx}\Big|_{x=x_0} \simeq \left[\frac{f(x + \Delta x, t) - f(x, t)}{\Delta x} \right] + O(\Delta x)$$

对于多个式子长度不一，但是式子结构相似，我们选择左对齐方式。

```

\begin{align*}
&\frac{\partial \theta}{\partial t}\Big|_{t=t_0} \approx \frac{\theta(t_0) - \theta(t - t_0)}{(t - t_0)} \\
&\frac{\partial \theta}{\partial t}\Big|_{t=t_0} \approx \frac{\theta(t_0) - \theta(t_{-1})}{\Delta t}
\end{align*}

```

$$\frac{\partial \theta}{\partial t}\Big|_{t=t_0} \approx \frac{\theta(t_0) - \theta(t - t_0)}{(t - t_0)}$$

$$\frac{\partial \theta}{\partial t}\Big|_{t=t_0} \approx \frac{\theta(t_0) - \theta(t_{-1})}{\Delta t}$$

```

\begin{align}
&\int_0^{\infty} \frac{\sin(x)}{x} \mathrm{d}x = \frac{\pi}{2} \\
&\int_0^{\infty} \frac{\sin(x)}{x} \frac{\sin(x/3)}{x/3} \mathrm{d}x = \frac{\pi}{2} \\
&\int_0^{\infty} \frac{\sin(x)}{x} \frac{\sin(x/3)}{x/3} \frac{\sin(x/5)}{x/5} \mathrm{d}x = \frac{\pi}{2}
\end{align}

```

```

& \int_0^{\infty} \frac{\sin (x)}{x} \frac{\sin (x / 3)}{x / 3} \frac{\sin (x / 5)}{x / 5} \frac{\sin (x / 7)}{x / 7} \mathrm{d} x=\frac{\pi}{2} \\
& \int_0^{\infty} \frac{\sin (x)}{x} \frac{\sin (x / 3)}{x / 3} \ldots \\
& \frac{\sin (x / 13)}{x / 13} \mathrm{d} x=\frac{\pi}{2} \\
\end{align}

```

$$\int_0^{\infty} \frac{\sin(x)}{x} dx = \frac{\pi}{2} \quad (1)$$

$$\int_0^{\infty} \frac{\sin(x)}{x} \frac{\sin(x/3)}{x/3} dx = \frac{\pi}{2} \quad (2)$$

$$\int_0^{\infty} \frac{\sin(x)}{x} \frac{\sin(x/3)}{x/3} \frac{\sin(x/5)}{x/5} dx = \frac{\pi}{2} \quad (3)$$

$$\int_0^{\infty} \frac{\sin(x)}{x} \frac{\sin(x/3)}{x/3} \frac{\sin(x/5)}{x/5} \frac{\sin(x/7)}{x/7} dx = \frac{\pi}{2} \quad (4)$$

$$\int_0^{\infty} \frac{\sin(x)}{x} \frac{\sin(x/3)}{x/3} \ldots \frac{\sin(x/13)}{x/13} dx = \frac{\pi}{2} \quad (5)$$

多个折行公式对齐:

```

\begin{align*}
(Z_2-Z_1) &= -\frac{R_d}{g_0} \bar{T}_v \ln \left( \frac{dp}{p} \right) \\
&= \frac{287 \times 288 \text{ K}}{9.81 \text{ m/s}^2} \ln \frac{1000 \text{ hPa}}{500 \text{ hPa}} \\
&= 5840.2419 \text{ km} \\
(Z_2-Z_1) &= -\frac{R_d}{g_0} \bar{T}_v \ln \left( \frac{dp}{p} \right) \\
&= \frac{287 \times 233 \text{ K}}{9.81 \text{ m/s}^2} \ln \frac{1000 \text{ hPa}}{500 \text{ hPa}} \\
&= 4724.9179 \text{ km}
\end{align*}

```

$$\begin{aligned}
(Z_2 - Z_1) &= -\frac{R_d}{g_0} \bar{T}_v \ln \left(\frac{dp}{p} \right) \\
&= \frac{287 \times 288 K}{9.81 m/s^2} \ln \frac{1000 hPa}{500 hPa} \\
&= 5840.2419 km \\
(Z_2 - Z_1) &= -\frac{R_d}{g_0} \bar{T}_v \ln \left(\frac{dp}{p} \right) \\
&= \frac{287 \times 233 K}{9.81 m/s^2} \ln \frac{1000 hPa}{500 hPa} \\
&= 4724.9179 km
\end{aligned}$$

三、多个式子居中对齐

独立的多个公式长短不一，或者投稿期刊排版要求等，有时候要求公式居中对齐，公式居中对齐我们通常采用 gather 数学公式环境来实现。

```

\begin{gather*}
\underbrace{\frac{DT}{Dt}}_{\text{Lagrangian Derivative}} = \\
\underbrace{\underbrace{\frac{\partial T}{\partial t}}_{\text{Local derivative}}}

```

```

+ \underbrace{\vec{V}\cdot \nabla T}_{\text{Advection Term}}_{\text{Euler
Derivative}}\backslash
\frac{DT}{Dt} = \frac{\partial T}{\partial t} + u\frac{\partial T}{\partial x} +
v\frac{\partial T}{\partial y} + w\frac{\partial T}{\partial z}
\end{gather*}

```

$$\underbrace{\frac{DT}{Dt}}_{\text{Lagrangian Derivative}} = \underbrace{\frac{\partial T}{\partial t} + \underbrace{\vec{V} \cdot \nabla T}_{\text{Advection Term}}}_{\text{Euler Derivative}}$$

$$\frac{DT}{Dt} = \frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} + w \frac{\partial T}{\partial z}$$

在居中对齐情况中，有时候遇见长式子折行等情况，通常会用 `aligned` 环境将整个小块包起来，这一小块内部可以设置对齐方式。

```

\begin{gather*}
\begin{aligned}
\hat{t}=t, \hat{x}=x, \hat{y}=\frac{y}{\sqrt{\epsilon}}
\end{aligned}
\begin{aligned}
\partial_t \rho + \partial_x (\rho u) &= \partial_t \rho + \partial_x (\rho u) \\
&= \partial_t \rho + (u_1 \partial_x + u_2 \partial_y) \rho \\
&= 0
\end{aligned}
\end{gather*}

```

$$\begin{aligned}
 \hat{t} &= t, \\
 \hat{x} &= x, \\
 \hat{y} &= \frac{y}{\sqrt{\epsilon}} \\
 \partial_t \rho + \nabla \cdot (\rho u) &= \partial_t \rho + (u \cdot \nabla) \rho + \rho (\nabla \cdot u) \\
 &= \partial_t \rho + (u_1 \partial_x + u_2 \partial_y) \rho \\
 &+ \rho (\partial_x u_1 + \partial_y u_2) \\
 &= 0
 \end{aligned}$$

四、方程组对齐

在方程组公式输入过程中，我们通常采用的数学环境是 `cases` 环境，值得注意的是，`cases` 环

境不能单独使用，需要套在 align、equation 等数学公式环境下才可以使用。

$$\begin{cases} \ddot{\theta} + \omega^2 \theta = 0 \\ \frac{d^2 \theta}{dt^2} + \omega^2 \theta = 0 \\ \theta(t) = A \cos(\omega t) + B \sin(\omega t) \end{cases}$$

```
\begin{equation*}
\begin{cases}
\ddot{\theta} + \omega^2 \theta = 0 \\
\frac{d^2 \theta}{dt^2} + \omega^2 \theta = 0 \\
\theta(t) = A \cos(\omega t) + B \sin(\omega t)
\end{cases}
\end{equation*}
```

$$\begin{cases} \ddot{\theta} + \omega^2 \theta = 0 \\ \frac{d^2 \theta}{dt^2} + \omega^2 \theta = 0 \\ \theta(t) = A \cos(\omega t) + B \sin(\omega t) \end{cases}$$

对于 cases 环境中的多个公式，我们还可以通过 “&” 来固定对齐位置，下面给出案列：

```
\begin{equation*}
\begin{cases}
\{\bm{Ax}\} \leq \{\bm{d}\}, & \{\bm{x}\} \in \{0, 1\} \\
\{\bm{Hu}\} \geq \{\bm{h}\}, & \{\bm{u}\} \in \{0, 1\}
\end{cases}
\end{equation*}
```

$$\begin{cases} \mathbf{Ax} \leq \mathbf{d}, & \mathbf{x} \in \{0, 1\} \\ \mathbf{Hu} \geq \mathbf{h}, & \mathbf{u} \in \{0, 1\} \end{cases}$$

我们在方程组输入过程中，有时候局部的方程组式子之间的间距过近（如分式、求和符号等），局部的调整可以在断行命令 “\\” 后紧跟 “[输入宽度如：20pt]” 命令来调整两个式子之间的行间距。

```
\begin{equation*}
\begin{cases}
f(t+t_0) &= f(t_0) + (t+t_0) \frac{\partial f}{\partial t} \Big|_{t=t_0} + \\
&\quad \frac{(t+t_0)^2}{2!} \frac{\partial^2 f}{\partial t^2} \Big|_{t=t_0} + \cdots \\
f(t+\Delta t) &= f(t_0) + \Delta t \frac{\partial f}{\partial t} \Big|_{t=t_0} + \\
&\quad \frac{(\Delta t)^2}{2!} \frac{\partial^2 f}{\partial t^2} \Big|_{t=t_0} + \\
&\quad \cdots
\end{cases}
\end{equation*}
```

$$\begin{cases} f(t+t_0) = f(t_0) + (t+t_0)\frac{\partial f}{\partial t}\Big|_{t=t_0} + \frac{(t+t_0)^2}{2!}\frac{\partial^2 f}{\partial t^2}\Big|_{t=t_0} + \dots \\ f(t+\Delta t) = f(t_0) + \Delta t\frac{\partial f}{\partial t}\Big|_{t=t_0} + \frac{(\Delta t)^2}{2!}\frac{\partial^2 f}{\partial t^2}\Big|_{t=t_0} + \dots \end{cases}$$

存在方程组长公式需要公式之间对齐、折行也要对齐时，我们可以考虑：

```
\begin{equation*}
```

```
\begin{cases}
```

```
T_{v,a} &= T_a(1+0.61q_a)\
```

```
&= 302(1+0.61 \times 24 \times 10^{-3})\
```

```
&= 306.42128 K\
```

```
T_{v,s} &= T_s(1+0.61q_s)\
```

```
&= 303(1+0.61 \times 5 \times 10^{-3})\
```

```
&= 303.92415 K
```

```
\end{cases}
```

```
\end{equation*}
```

$$\begin{cases} T_{v,a} = T_a(1 + 0.61q_a) \\ = 302(1 + 0.61 \times 24 \times 10^{-3}) \\ = 306.42128K \\ T_{v,s} = T_s(1 + 0.61q_s) \\ = 303(1 + 0.61 \times 5 \times 10^{-3}) \\ = 303.92415K \end{cases}$$

在 cases 公式环境中，默认方程组括号是在左边，有时候我们需要用到括号在右边括起来，这时候我们可以考虑使用 rcases 环境。

```
\begin{align*}
```

```
\begin{rcases}
```

```
f(x+\Delta x) &= f(x) + \Delta x\frac{\partial f}{\partial x}\Big|_{x=x_0}\
```

```
&+ \frac{(\Delta x)^2}{2!}\frac{\partial^2 f}{\partial x^2}\Big|_{x=x_0}\
```

```
&+ \frac{(\Delta x)^3}{3!}\frac{\partial^3 f}{\partial x^3}\Big|_{x=x_0}+\quad\cdots
```

```
\end{rcases}\
```

```
\begin{rcases}
```

```
f(x-\Delta x) &= f(x) - \Delta x\frac{\partial f}{\partial x}\Big|_{x=x_0}\
```

```
&+ \frac{(\Delta x)^2}{2!}\frac{\partial^2 f}{\partial x^2}\Big|_{x=x_0}\
```

```
&- \frac{(\Delta x)^3}{3!}\frac{\partial^3 f}{\partial x^3}\Big|_{x=x_0}+\quad\cdots
```

```
\end{rcases}
```

```
\end{align*}
```

$$\left. \begin{aligned}
f(x + \Delta x) &= f(x) + \Delta x \frac{\partial f}{\partial x} \Big|_{x=x_0} \\
&+ \frac{(\Delta x)^2}{2!} \frac{\partial^2 f}{\partial x^2} \Big|_{x=x_0} \\
&+ \frac{(\Delta x)^3}{3!} \frac{\partial^3 f}{\partial x^3} \Big|_{x=x_0} + \dots
\end{aligned} \right\}$$

$$\left. \begin{aligned}
f(x - \Delta x) &= f(x) - \Delta x \frac{\partial f}{\partial x} \Big|_{x=x_0} \\
&+ \frac{(\Delta x)^2}{2!} \frac{\partial^2 f}{\partial x^2} \Big|_{x=x_0} \\
&- \frac{(\Delta x)^3}{3!} \frac{\partial^3 f}{\partial x^3} \Big|_{x=x_0} + \dots
\end{aligned} \right\}$$